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## RESEARCH ARTICLE

# Improved Artificial Bee Colony Algorithm Based on Multi-Strategy Synthesis for UAV Path Planning

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**ABSTRACT** Path planning is the key for unmanned aerial vehicle (UAV) to perform tasks efficiently, which needs to quickly obtain the optimal path in the complex environment. To solve the path planning problem in the complex urban environment, an improved artificial bee colony algorithm based on multi-strategy synthesis (IABC) is proposed to generate the appropriate path for the UAV. The IABC is based on the hybrid mechanism of chaotic mapping and Pareto principle to initialize the population, so as to fully search the solution space and provide an approximate optimal flight path for UAV. Meanwhile, in order to balance exploration and development, two new search equations are designed to generate candidate solutions, which provide more superior flight paths for UAVs. In addition, tangent random evolution mechanism is added to enhance the strategy of updating the algorithm offspring, maximize the quality of flight path generation, and effectively solve the path planning problem. Aiming at the complex urban environment and the multi-constraint optimization problem of UAV flight, the IABC is combined with cubic spline interpolation to generate a smooth path to meet the UAV flight maneuver characteristics. The improved algorithm proposed in this paper has been compared with Particle Swarm Optimization (PSO), Artificial Bee Colony (ABC), Search Artificial Bee Colony (SABC) and Gbest Artificial Bee Colony (GABC). The algorithm proposed in this paper has good feasibility and effectiveness in solving the UAV path planning problem.

**INDEX TERMS** Artificial bee colony algorithm (ABC), path planning, unmanned aerial vehicle (UAV), cubic spline interpolation.

## I. INTRODUCTION

Unmanned Aerial Vehicle (UAV) is a comparatively small and portable remote control vehicle. In virtue of its advantages of low cost and many applicable scenarios, UAV is widely used in the atmospheric exploration industry, the power industry and other industries [1], [2]. UAV path planning is the pivotal technology of UAV mission planning system. It generally refers to series of optimization problems after determining the take-off position and goal mission of UAV. According to the mobility of UAV, geographical

environment threat and constraint conditions, an optimum or suboptimal flight path is planned [3]. The pros and cons of the planning track directly affects whether the UAV can successfully complete the task. Therefore, three-dimensional (3D) trajectory planning technology has become the key to the development of UAV. In this paper, we propose a method of UAV path planning in urban 3D complex environment.

In recent decades, group intelligence optimization algorithms have developed roundly. Such as Genetic Algorithm (GA) [4], Ant Colony Optimization (ACO) [5], Particle Swarm Optimization (PSO) [6] and so on, these algorithms can solve the problem of flight planning in complex environment. For example, the idea of ACO algorithm is applied

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to the coverage path planning problem of heterogeneous UAV, which reduces the task completion time in large-scale collaborative search system in [7]. Reference [8] combined differential evolution with ACO to reasonably allocate pheromones left by ants in each path, in order that guide ants to find the optimal path. The PSO was studied from the perspective of quantum mechanics in [9]. The concepts of Gbest and Mbest are introduced to make the update of particle position and velocity random. Reference [10] introduced the network construction operator in the PSO algorithm. After updating the position of the population particles, the operator is applied to each particle to overcome the problem of PSO algorithm falling into local optimum and slow convergence speed in the later stage.

Artificial Bee Colony (ABC) [11] is a group intelligent optimization proposed by Turkish scholar Karaboga in 2005. It is a biomimetic algorithm to solve practical engineering optimization problems by simulating the behavior of bee colony to find excellent honey sources. Compared with the afore-said example algorithm, ABC algorithm has low complexity and strong adaptability. Its bionic structure makes it have the advantages of realization and flexibility in solving practical problems. The experimental results show that ABC algorithm performs very well in solving optimization problems [12], [13]. At present, ABC has been applied in numerical function optimization [14], pipeline scheduling [15], mobile communication system [16] and path planning [17]. Such as, Reference [18] improved the ABC algorithm by using the global optimum strategy, introduced the adaptive parameters into the search equation, and improved the ability of the algorithm to seek global optimum. Reference [19] modified the new honey source search formula to find out the best honey source near the honey source for exploitation and improve the local optimization ability of the algorithm. Reference [20] combined ABC with GA, and used crossover and exchange mechanism in GA to improve the update formula of bee, so that it can jump out of local optimal solution. The concept of QE-ABC is proposed to improve the randomness of initial population and individual local exploration ability in [21].

Despite these improved algorithms [18], [19], [20], [21] have been proved to be effective in solving optimization problems, there are still the following shortcomings: The improved method is limited to partial optimization of the algorithm, and does not consider the characteristics of the bee population as a whole. Although it produces better optimization results in individual stages, when the dimension of the optimization problem increases, the performance of the ABC algorithm will be reduced like other swarm intelligence algorithms. However, in complex flight environments, UAV path planning problems are often high-dimensional. In addition, usually in the path planning stage, if the kinematic constraints are not considered, the path planned by the algorithm cannot fly in the actual situation [22], [23], [24]. There are many unprocessed track turning points and lack of continuous smoothness. In the case of sudden stop and

sharp turn, due to the special flight maneuver characteristics of the drone, it is easy to cause damage to related components. At present, the commonly used method is to fit the flight path generated by the algorithm [25] [26], [27], but this method increases the workload of the project.

Track planning in urban of the 3D complex environment not only contains a great quantity of obstacles, but also has multiple-flying constraints. The problem is difficult to solve, and the ABC algorithm still has problems such as slow search speed, reduced population diversity and falling into local optimal solution when generating the best flight path. Aiming at the path planning problem of ABC algorithm in 3D complex environment, this paper proposes a multi-strategy comprehensive improved artificial bee colony optimization algorithm. The difference of this paper is that the smooth path is directly generated by the method of cubic spline interpolation coding [28]. The track curve generated by this method is smoother, which can ensure that the UAV has better dynamic characteristics when it stops and turns abruptly. From the perspective of overall optimization, the corresponding overcoming strategies are designed for the stages where ABC algorithm is prone to defects, so as to ensure that the flight trajectory generated in static complex space is efficient and smooth. The main contributions of this paper are as follows:

(1) According to the characteristics of complex urban environment and multi-constraint conditions of UAV flight, the coding method based on the definition of path nodes based on the cubic spline interpolation method is adopted to construct the fitness function of UAV without collision, which effectively avoids the generation of non-flying paths and makes the generated flight paths smoother, which is in line with the characteristics of UAV flight.

(2) Tent chaotic map and Pareto principle are used to initialize the population, which improves the diversity and quality of bee population, and improves the efficiency of the algorithm. Through the competitive elimination mechanism between bees, the tangent random population evolution strategy is further designed, which not only ensures the efficient updating of the population, but also makes the optimization mechanism of the algorithm more strategic.

(3) In view of the simple search method of the traditional ABC algorithm, the guided search strategy is added to the bee-harvesting stage. The bee-harvesting learns the experience of the excellent individuals of the father generation, which can not only achieve faster convergence, but also improve the accuracy of the algorithm. The step size factor is introduced in the reconnaissance bee stage to make the algorithm easier to jump out of the local extremum and improve the search efficiency of the algorithm.

The structure of this paper is organized as follows. Section 2 describes the problem formula, including constraints and environmental modeling. In Section 3, IABC is proposed and the improvement measures are described in detail. Section 4 combines the problem with the cubic spline interpolation method, constructs the fitness function,

and gives the path planning flow chart. Section 5 provides simulation results of several algorithms. Finally, Section 6 summarizes the research.

## II. 3D ENVIRONMENT MODELING OF UAVS

### A. FLIGHT SPACE

UAVs perform flight tasks in 3D space, which span a wide range of space, and the environment is complex. Moreover, UAVs constrained by many conditions. Therefore, path planning needs to balance the search time, path cost and computational complexity to content the requirements of practical engineering. The goal of trajectory planning is to find a feasible path to the target point under complex environment and various constraints. In the environmental model, the position of the track point is represented by 3D coordinates  $(x_i, y_i, z_i) (i = 1, 2, \dots, Ns)$ , and the UAV flight space is express as:

$$\{(x_i, y_i, z_i) | x_{min} \leq x_i \leq x_{max}, y_{min} \leq y_i \leq y_{max}, z_{min} \leq z_i \leq z_{max}\} \quad (1)$$

where,  $x_{min}, x_{max}, y_{min}, y_{max}, z_{min}, z_{max}$  indicates flight space boundary, respectively.

### B. CONSTRAINT CONDITIONS

#### 1) DISTANCE COST CONSTRAINT

In the process of UAV carrying out flight tasks, the track with shorter distance ( $M_{first}, M_2, M_3, \dots, M_{last}$ ) has better cost performance than the track with longer distance. The flight cost is reduced by shortening the distance, reducing the fuel consumption of the uav and reducing the flight time. Considering the limited fuel carried by the UAVS themself, the flight distance should be controlled in a certain limit, equation (2) can be used to characterize the distance cost constraint.

$$G = \sum_{i=0}^N f_i \quad (2)$$

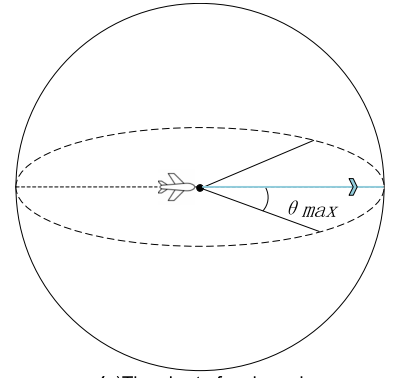
$$f_i = \sqrt{(x_{M_{i+1}} - x_{M_i})^2 + (y_{M_{i+1}} - y_{M_i})^2 + (z_{M_{i+1}} - z_{M_i})^2} \quad (3)$$

where,  $G (G \leq G_{max})$  represents distance cost,  $N$  denotes the number of track segments forming a complete track,  $f_i$  shows the distance cost of track  $i$ .

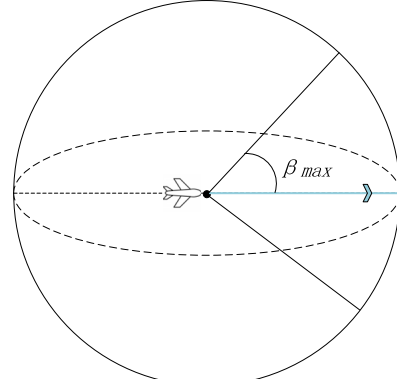
#### 2) FLIGHT HEIGHT CONSTRAINT

Air pressure, wind speed and other factors have a huge impact on the flight of UAV. considering the flight safety of UAV, the flight height should be controlled in a certain range, and the flight height should not be too high or too low. The terrain height  $T_{map}$  of the environmental model at a track point  $(x_{M_i}, y_{M_i}, z_{M_i})$  is set to be,  $f_h$  represents the minimum flight altitude,  $F$  is defined as minimum flight altitude adjustable information.

$$F = \begin{cases} 0, & f_h \leq T_{map} \\ f_h - T_{map}, & otherwise \end{cases} \quad (4)$$



(a) The chart of crab angle



(b) The chart of pitch angle

FIGURE 1. Charts of crab angle and pitch angle.

#### 3) MAXIMUM CRAB ANGLE CONSTRAINT

Accompanied by flight, the UAVs need to adjust its flight attitude in time, which are restricted by its mobility. Only when the crab angle is less than the maximum crab angle, the uav can maintain a stable flight attitude. Let the vector of a track segment and its adjacent track segment be  $p_i = (x_{M_{i+1}} - x_{M_i}, y_{M_{i+1}} - y_{M_i}, z_{M_{i+1}} - z_{M_i})$ , Maximum carb angle should satisfy constraints:

$$\cos \theta_{max} \leq \frac{p_{i+1}^T \times p_i}{\sqrt{(p_{i+1})^2 \times (p_i)^2}} \quad (5)$$

#### 4) MAXIMUM PITCH ANGLE CONSTRAINT

The factors affecting the pitch angle of UAVs include wind direction, climates and its own mobility. This constraint limits the UAVs to ride-up or slide-down in a specific angle range. The maximum pitch angle can be described by the mathematical expression:

$$\tan \beta_{max} \geq \frac{|z_{M_{i+1}} - z_{M_i}|}{p_i} \quad (6)$$

### C. GEOGRAPHY MODEL

In the 3D environment, the UAVs should consider the spatial location of the threat source, the radius of the threat source and other restrictions. The threat area is geographical obstacles, enemy radar and so on. This paper is based on the 3D space model established by the city, and the topology of

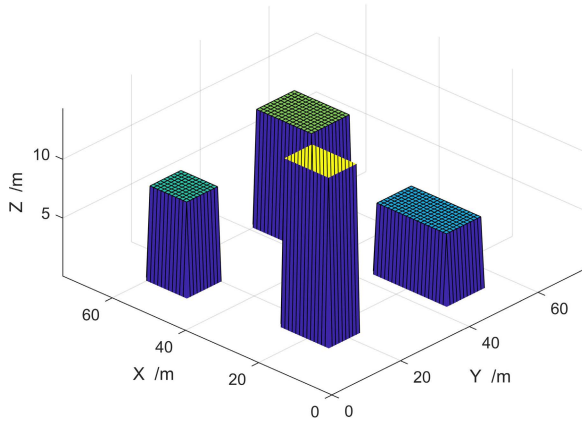


FIGURE 2. Urban environment simulation model.

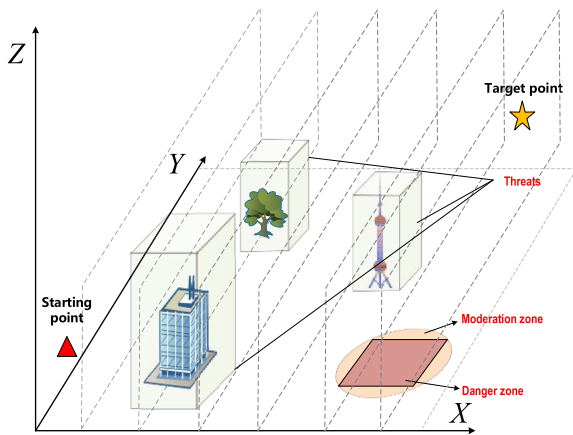


FIGURE 3. Environmental threat model.

the tall buildings and other environmental constructions in the urban space is a rectangular body with different rules, as shown in Fig. 2. For the urban dangerous obstacles in the global map environment, we divide the danger zone, the moderation zone and the safety zone into three levels. In the simulation environment, the path is required to pass the barrier in the moderation zone and the safety zone. In Fig. 3, you can clearly see the plane division of the danger zone and the moderation zone, the obstacles are represented by  $[0,1]$  model, 0 represents the feasible area, 1 represents the obstacle. Then the obstacle risk of the path node is:

$$M_b = \begin{cases} 0, & M_{building}(x,y,z) = 0 \\ \inf, & M_{building}(x,y,z) = 1 \end{cases} \quad (7)$$

where,  $M_{building}$  is the index matrix of obstacles. By modeling the area where the path section is located by using the representation of probability, it is easy to obtain the probability distribution of being blocked by buildings, as a means to select the optimal path traffic scheme.

### III. UAV PATH PLANNING BASED ON IABC ALGORITHM

#### A. INTRODUCTION OF ABC ALGORITHM

In the basic artificial bee colony algorithm, each bee is regarded as a massless and sizeless particle, which represents

a feasible solution in the solution space. For different fitness functions, each bee has corresponding values. By comparing the fitness function values, the optimal individual in the current population is determined. Then, according to the exploration, evolution and information sharing of other bees in the population, the global optimal solution is determined. In the ABC algorithm, the honey-harvesting system of bee colony is mainly composed of honey source, collecting bee, following bee and reconnoitering bee. The collecting bee and following bee account for half of the total population, respectively. The collecting bee is responsible for finding the honey source and conducting field search in the initialization phase, and returns to the hive to dance and share the honey source information with the following bee. The following bee receives the information shared by the collecting bee in the hive, and conducts the honey-harvesting task near the honey source. The reconnoitering bee conducts the global exploration after the honey source is abandoned, and finds a new honey source to replace the original honey source.

#### 1) INITIALIZATION OF ABC

Initializing the parameters of ABC algorithm, the parameters include the number of honey sources  $N_s$ , the number of collectors and followers  $N_e$ , the maximum number of iterations  $iter_{max}$ , and the number threshold of local optimum  $limit$ . Randomly generating  $i$  ( $i = 1, 2, \dots, N_s$ ) honey source locations in the global:

$$x_{id}^0 = Min_d + rand(0, 1) \times (Max_d - Min_d) \quad (8)$$

where,  $d = 1, 2, \dots, D$  is the function dimension,  $Min_d$  and  $Max_d$  represent the lower and upper limits of the  $d$ th dimension respectively.

#### 2) COLLECTING BEE STAGE

In the stage of collecting bees, collecting bees use the following formula to find new sources of honey:

$$x_{id}^{t+1} = x_{id}^t + \ell \times (x_{id}^t - x_{kd}^t) \quad (9)$$

where,  $k = 1, 2, \dots, Ne$  and  $k \neq i$ .  $\ell$  means a random number with an interval of  $[-1,1]$  and determines the extent to which the current individual is affected by other individuals. The new honey source is generated by Refer to (9), and the fitness function is calculated. According to the greedy selection criterion, the new and old honey sources are selected optimally. The fitness function is:

$$fitness_i = \begin{cases} 1 / (1 + fit_i), & fit_i \geq 0 \\ 1 + |fit_i|, & fit_i < 0 \end{cases} \quad (10)$$

where,  $fit_i$  is the objective function value of the  $i$ th honey source.

#### 3) FOLLOWING BEE STAGE

Followers receive the honey source information shared by collectors in the honeycomb dance area, and calculate the

probability according to (11).

$$Pc_i = \frac{fitness_i}{\sum_{i=0}^{Ne} fitness_i} \quad (11)$$

The roulette strategy is used to select the honey source development to ensure that the honey source with high fitness value is more likely to be mined. The followers search for the optimal honey source in the field refer to (9) for exploitation. If the honey source changes, the value is set *limit* to zero. On the contrary, *limit* + 1.

#### 4) RECONNOITERING BEE STAGE

If a honey source is mined many times without obtaining a new honey source, that is determined that the it does not have an optimal solution. The maximum threshold is setting *limit*. When it exceeds the threshold, the honey source is abandoned, and the scouter stage is started. Refer to (8), thereconnoitering bee randomly generates a new honey source, and the value is set *limit* to zero.

### B. FEATURES OF ALGORITHM PLANNING

The ABC algorithm is applied to the path planning of UAV, and the idea of bee seeking the optimal honey source process is applied to the path planning. The algorithm is modified to be suitable for solving such problems. The path planning problem based on ABC algorithm is shown in Table 1.

**TABLE 1.** Correspondence of flight trajectory planning based on ABC algorithm.

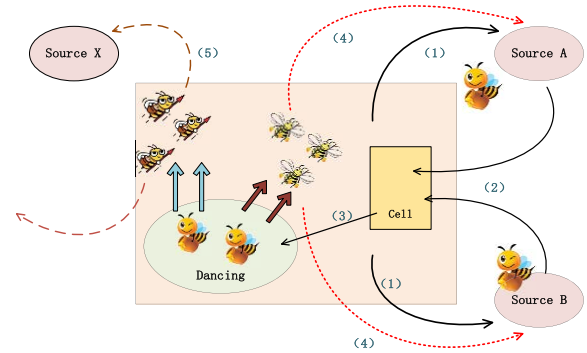
| ABC algorithm                          | Path planning                                 |
|----------------------------------------|-----------------------------------------------|
| Honey picking behavior of bee colony   | routing optimization problem                  |
| Honey source location                  | Coordinate position of feasible path solution |
| The size of honey income               | Quality of path feasible solutions            |
| Finding honey source and picking speed | Speed of solving path optimization            |
| Maximum profit of honey source         | The cost of optimal path                      |

### C. IMPROVEMENT STRATEGY OF IABC ALGORITHM

The same as other intelligent algorithms, there is always a contradiction between the diversity of the population and the deep mining ability of the algorithm in the optimization process of the ABC algorithm. The purpose of improving the standard ABC algorithm is to strengthen the local search ability of the algorithm and maintain the diversity of the population, so as to prevent premature convergence of the algorithm. The improvement in this paper is also based on this idea, and the specific improvement strategies are introduced in detail below.

#### 1) INITIALIZATION STRATEGY OF TENT CHAOTIC ELITE POPULATION

The ABC algorithm stochastically generates the honey source position in the population initialization phase, and usually



**FIGURE 4.** Flow process chart of ABC algorithm for honey-harvesting.

uses the randomly generated data as the information of population initialization, which is difficult to ensure the diversity of the population, resulting in poor optimization results of the algorithm and affecting the optimization speed. Chaos is a common nonlinear phenomenon in nature. Chaos optimization search has the characteristics of randomness, regularity and global ergodicity. In the initial stage of population, it can maintain population diversity and improve the global search ability. At present, logistic chaotic mapping is used in the optimization stage of population initialization in [29], [30], and [31], but the mapping has a high value rate in regions (0,0.1) and (0.9,1) in Fig.5. In contrast, the ergodic inhomogeneity of logistic map will lead to a slower convergence speed of the algorithm, while the value rate of tent chaos in each interval is almost equal [32]. Therefore, this paper will use tent chaotic map to initialize the population.

The expression of tent chaotic:

$$x_{i+1,d} = \begin{cases} x_{id}/\mu, & x_{id} \in (0, \mu] \\ (1 - x_{id})/(1 - \mu), & x_{id} \in (\mu, 1] \end{cases} \quad (12)$$

When  $\mu = 1/2$ , Tent mapping has the most typical form, and the honey source position generated by population initialization has a uniform distribution. In this paper, the reference formula of  $\mu = 1/2$  is as follows:

$$x_{i+1,d} = \begin{cases} 2x_{id}, & x_{id} \in (0, 0.5] \\ 2(1 - x_{id}), & x_{id} \in (0.5, 1] \end{cases} \quad (13)$$

Namely  $x_{i+1,d} = (2x_{id}) \bmod 1$ .

In order to generate abundance of effective honey source locations in the solution space, the number of initialized populations should not be too small-scale. However, excessive populations will generate huge data calculations, resulting in slow operation of the algorithm. Therefore, after the Tent chaotic map initializes the population, it is combined with the idea of Pareto principle (80/20 rule) [33], the rule is used to find out 20 % of the better honey source in all populations as the elite group, and the elite group is used as the iterative group, which reduces the data redundancy and improves the operation efficiency of the algorithm.

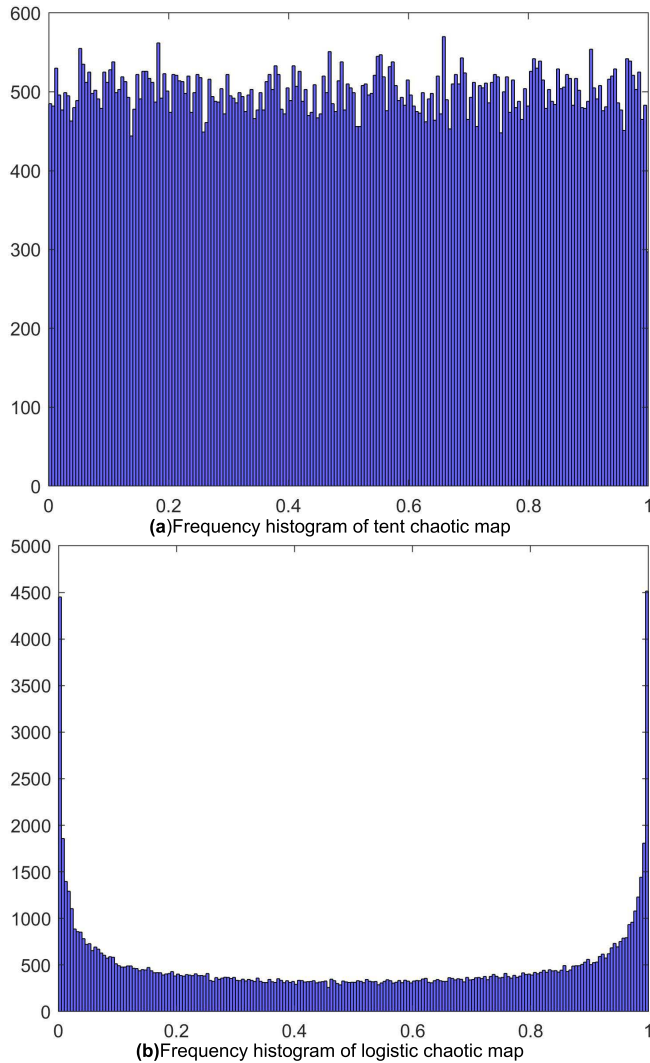


FIGURE 5. Frequency histogram of two chaotic maps.

## 2) GUIDING SEARCH STRATEGY

In the improvement of swarm intelligence algorithm, global exploration and local mining are necessary measures. Exploration refers to the ability to find the global optimal solution in the unknown solution space, and mining refers to the ability to search the local optimal solution in the known solution space. With an eye to make the algorithm get better optimization, these two abilities should be balanced.

The bee search equation described in equation (9) is based on the old solution, and is affected by another random solution to generate a new solution. The new solution is full of randomness, which is easy to mislead the follower bee to search in the direction of the suboptimal solution and increase the running time of the algorithm. In order to make the algorithm more efficient, we use the following formula as the collector search equation.

$$x_{id}^{t+1} = x_{id}^t + \ell \times (x_{id}^t - x_{kd}^t) + \psi \times (x_{best,d}^t - x_{id}^t) \quad (14)$$

where,  $\psi$  is a nonnegative constant,  $x_{best,d}^t$  is the optimal individual of the  $t$ th generation population in  $d$  dimensional space. According to the above formula, the newly generated candidate solution is close to the global optimal solution, which increases the utilization rate of the algorithm. The value of  $\psi$  should not be too large or too small. The value of  $\psi$  is too large to make the new solution jump into the local optimal trap quickly. If the value of  $\psi$  is too small, the influence of  $x_{best,d}^t$  on the new solution can be ignored.

## 3) LARGE-STEP EXPLORATION STRATEGY

The significance of following bee role is to jump out of local extremum in time when the solution falls into local optimum. Due to the complexity of the solution space of the simulation environment, it is difficult to obtain a better result than the current solution only by random reconnaissance, and there is a great probability that it does not jump out of the local optimum. Therefore, a large step search method is designed for scout bee to explore the optimal solution.

$$x_{id}^{t+1} = x_{id}^t + \frac{Map_d}{S} * h \quad (15)$$

where,  $S$  represents step size factor and  $\eta$  represents weight factor. When the *limit* value of a certain honey source exceeds the threshold, no near better honey source is found. At this time, considering that it has fallen into local optimum, the role of bees is converted into reconnaissance bees to find new honey sources.

## 4) TANGENT STOCHASTIC EVOLUTION STRATEGY

The efficient evolutionary mechanism can keep the swarm energetic, and the competition and information exchange among individuals can maximize the exploration potential of the swarm. In the optimization process, the bee colony uses structured and random information to meet the target individuals to obtain the maximum likelihood of survival, and continuously enhance the competitiveness of the entire swarm. Therefore, a tangent random evolution strategy based on the survival of the fittest is proposed to improve the quality of the offspring of the bee colony.

The survival of the fittest strategy makes those bees who fall into local optimum and are in poor honey source positions no longer participate in the search, and abandon inferior honey sources in advance. The survival of the fittest strategy measures the advantages and disadvantages of the honey source by the fitness value of the honey source position. Therefore, after the end of each generation of search, the new generation of individuals are arranged in ascending order according to the fitness value, and the  $m$  honey source with larger fitness value are eliminated, and the honey source with the same number of eliminated honey sources is generated to maintain the stability of the population. If it is too small, the number of new solutions in the algorithm is reduced, the diversity of the algorithm is reduced, and the development ability is weakened. However, if it is too large, the number of new solutions in the algorithm is increased, and the randomness of the algorithm is enhanced, which leads to the

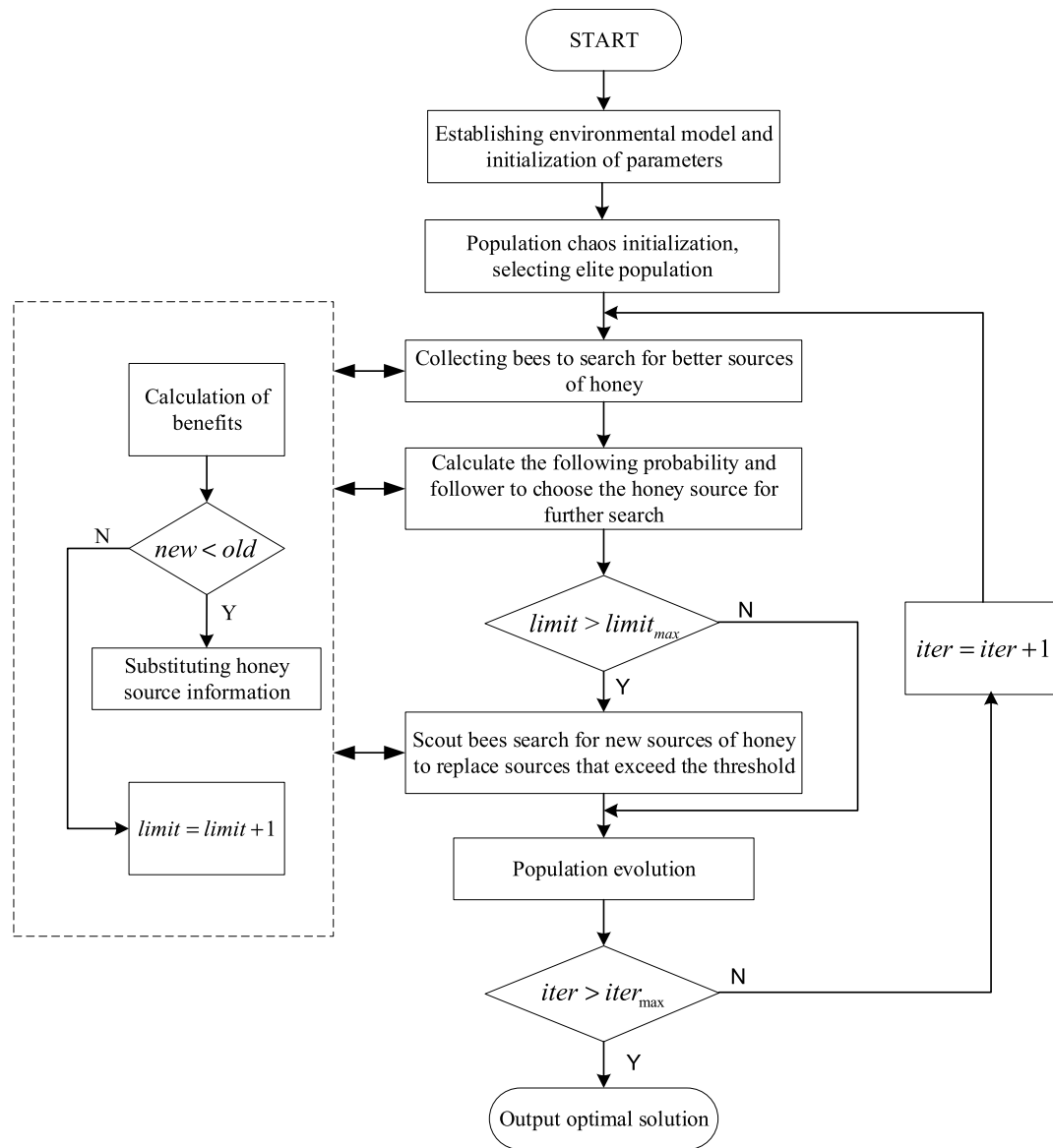


FIGURE 6. Flow diagram of the IABC algorithm.

TABLE 2. Information of threats.

| Designation | Number of obstacles | Feature [D,W,H] |            |            |
|-------------|---------------------|-----------------|------------|------------|
| Case 1      | 4                   | [13,9,15]       | [10,10,8]  | [16,12,10] |
|             |                     | [20,11,6]       |            |            |
| Case 2      | 6                   | [16,9,18]       | [10,9,15]  | [13,8,8]   |
|             |                     | [16,8,10]       | [15,11,6]  | [16,8,10]  |
| Case 3      | 12                  | [7,6,8]         | [11,10,12] | [11,10,6]  |
|             |                     | [15,11,20]      | [9,6,9]    | [11,8,11]  |
|             |                     | [10,9,14]       | [14,6,10]  | [13,8,5]   |

slow convergence speed of the algorithm. Therefore, in order to ensure the balance between accuracy and convergence speed of the algorithm. Deeming:

$$m = [Ne/\omega, Ne/(\omega \cdot 0.618)] \quad (16)$$

where,  $\omega$  is the updating scale factor,  $Ne$  indicates the population quantity,  $m$  satisfies the golden section ratio.

The solution accuracy and convergence efficiency of population should be guaranteed during evolution, a tangent

**TABLE 3.** Initial parameter information of algorithms.

| Algorithm | Main parameter                           | Value  |
|-----------|------------------------------------------|--------|
| PSO       | Particle number $M$                      | 100    |
|           | Social factor $C_1$                      | 2      |
|           | Cognition factor $C_2$                   | 2      |
|           | Inertance factor $w$                     | 1.2    |
| ABC       | Number of bees $Ns$                      | 100    |
|           | Threshold limit $limit$                  | 5      |
|           | Function allocation coefficient $\varpi$ | 0.5    |
| SABC      | Number of bees $Ns$                      | 100    |
|           | Threshold limit $limit$                  | 5      |
|           | Function allocation coefficient $\varpi$ | 0.5    |
|           | Random search factor $\kappa$            | (-1,1) |
| GABC      | Number of bees $Ns$                      | 100    |
|           | Threshold limit $limit$                  | 5      |
|           | Function allocation coefficient $\varpi$ | 0.5    |
|           | Gbest impact factors $A$                 | 3      |
|           | Random search constant $C$               | 1.5    |
| IABC      | Number of bees $Ns$                      | 100    |
|           | Threshold limit $limit$                  | 5      |
|           | Function allocation coefficient $\varpi$ | 0.5    |
|           | Guiding factor $\psi$                    | 2      |
|           | step length coefficient $S$              | 4      |
|           | Update proportion $\omega$               | 1.6    |
|           | controllable parameters $\sigma$         | 2      |

stochastic population evolution strategy is proposed, as shown in (17).

$$x_{id}^{t+1} = \begin{cases} x_{id}^t & i \leq m \\ x_{best,d}^t + \sigma \times \tan((r - 0.5) \times \pi) & otherwise \end{cases} \quad (17)$$

Which,  $\gamma$  is a uniformly distributed random number on  $[0,1]$ ,  $x_{best,d}^t$  is the best fitness individual for the  $t$  th generation of the population, and  $\sigma$  is the controllable factor. The value of  $\tan((r - 0.5) \times \pi)$  has two distribution characteristics: 1) the function value is mainly distributed around 0; 2) There are some numbers with large values. According to its distribution characteristics, the bee searches near the current optimal honey source location. Formula enhances the local exploitation ability of the algorithm, accelerates the convergence speed of the algorithm, and makes it easier for the algorithm to jump out of the local optimum as to maintaining the population diversity and making the evolution mechanism of the ABC algorithm more strategic, rather than just blind random exploration. The quality of the algorithm is enhanced.

#### IV. PATH PLANNING OF UAVS BASED ON CUBIC SPLINE INTERPOLATION

In order to meet the special flight maneuvering characteristics of the UAV, the flight path generated by the algorithm needs to be smoothed. The widespread method is to fit the generated flight path curve, which increases the workload of the project. The cubic spline interpolation is a piecewise interpolation method, which forms a smooth curve through a series of interpolation point intervals based on cubic polynomials. The UAV flight path curve fitted by the cubic spline interpolation method is smoother, and there is no need to curve fit the path planned by the algorithm. It can not only ensure that the UAV has better dynamic characteristics during emergency stop or emergency turn, but also reduce the redundancy of the path generation process. Therefore, this paper uses cubic spline interpolation curve to construct and represent UAV flight path.

##### A. CUBIC SPLINE INTERPOLATION

Let  $\Delta$  be a partition of interval  $[a,b]$ , if function  $S(x)$  satisfies:

- (1)  $S(x) \in C^2[a, b]$
- (2)  $S(x_i) = f(x_i) \quad i = 0, 1, 2, \dots, n;$

(3)  $S(x)$  is cubic polynomial on every subinterval  $[x_i, x_{i+1}]$  ( $i = 0, 1, 2, \dots, n-1$ ). Then  $S(x)$  is called a cubic spline function for partition  $\Delta$ .

If a function value is given on node  $x_i$ , and  $S(x_i) = y_i$ , then  $S(x)$  is called a cubic spline interpolation function. To determine the only, three common boundary conditions need to be added:

- 1)  $S'(a) = y'_0, S'(b) = y'_n$ ;
- 2)  $S''(a) = y''_0, S''(b) = y''_n$ ;
- 3)  $S(a+0) = S(b-0), S'(a+0) = S'(b-0), S''(a+0) = S''(b-0)$ .

## B. CODE DESIGN

Cubic spline interpolation is a piecewise interpolation method. The splines between segments are different. The whole spline curve is first-order continuous and the path nodes are second-order continuous. Therefore, the number of path nodes represents the commutation times of the whole path. Although the environmental conditions are very complex, it is generally only 3-5 times to avoid all obstacles. The location of path nodes is selected according to environmental changes.

In this paper, all nodes on a path are considered as the code of a bee, that is to say, the 3D coordinates of  $n$  path nodes on a path constitute  $n$ -3D coordinates of the bee individual. Assuming that the coordinates of  $n$  path nodes have been determined, respectively,  $r_1 = (x_{r_1}, y_{r_1}, z_{r_1}), \dots, r_n = (x_{r_n}, y_{r_n}, z_{r_n})$ , the starting point and end point coordinates of the path are  $(x_s, y_s, z_s)$  and  $(x_t, y_t, z_t)$ . respectively, on the interval  $[x_s, x_{r_1}, x_{r_2}, \dots, x_{r_n}, x_t]$ ,  $[y_s, y_{r_1}, y_{r_2}, \dots, y_{r_n}, y_t]$  and  $[z_s, z_{r_1}, z_{r_2}, \dots, z_{r_n}, z_t]$ , the coordinates of interpolation points are  $(x_1^S, x_2^S, \dots, x_m^S), (y_1^S, y_2^S, \dots, y_m^S)$  and  $(z_1^S, z_2^S, \dots, z_m^S)$ , and by cubic spline interpolation. Then the path point and interpolation point as well as the starting point and end point connection curve is the path of UAV planning.

## C. FITNESS FUNCTION AND COLLISION DETECTION

The flight trajectory of UAV cannot collide with obstacles, and the path length is required to be as short as possible. The purpose of collision detection is to detect whether the planned path passes through obstacles. The curve processed by cubic spline interpolation is segmented according to the path node. Each node produces  $\{(x, y) | (x_{r_i}, y_{r_i}), (x_1^S, y_1^S), \dots, (x_m^S, y_m^S)\}$  and  $\{z | z_{r_i}, z_1^S, \dots, z_m^S\}$  sets, and the two sets constitute a point in the path curve.

This paper defines obstacles as ternary function  $F = aX + bY + cZ$ . ( $a, b, c$ ) represents weight interpolates. The function  $F$  data is interpolated, and the interpolation function value corresponding to the set  $\{(x, y)\}$  value is obtained. The function value is compared with the set  $\{(z)\}$  to determine whether the path passes through the obstacle. The fitness function constructed in this paper is:

$$z = L \otimes \varpi \quad (18)$$

$$\varpi = \varpi_1 f_1 + \varpi_2 f_2 + \varpi_3 f_3 \quad (19)$$

$$L = \sum_{i=1}^n \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2 + (z_{i+1} - z_i)^2} \quad (20)$$

where,  $f_1, f_2, f_3$  respectively represents the cost of flight altitude, maximum yaw angle and maximum elevation angle. And fulfils  $\varpi_1 + \varpi_2 + \varpi_3 = 1$ .

## D. IABC ALGORITHM FLOW

**Step 1:** Establish three-dimensional space environment model and build fitness function.

**Step 2:** The maximum number of iterations of the algorithm is  $iter_{max}$ , the population size  $Ns$ , the number of elite bees is  $Ne$ , and the maximum threshold is  $limit$ . According to the specific environment, the number of path nodes is  $n$ , the number of interpolation points is  $m$ , and the starting point and ending point of UAV are  $(x_s, y_s, z_s)$  and  $(x_t, y_t, z_t)$ .

**Step 3:** Initialize the population according to equation (13), arrange the population according to equation (18), and determine the elite population.

**Step 4:** The iterative search was carried out to explore the better honey source in the honey source position according to the equation (14).

**Step 5:** The fitness value is calculated according to equation (18) and the results are compared with the honey source before the search. If the fitness value is smaller than the previous fitness value, the honey source information is replaced, otherwise  $limit + 1$ .

**Step 6:** Follower according to equation (14) to pick up honey. And update information according to Step5.

**Step 7:** When the value exceeds the maximum threshold, a new honey source position is generated according to equation (15), and the value is set to zero, and the information is updated according to Step5.

**Step 8:** Population evolution according to equation (17).

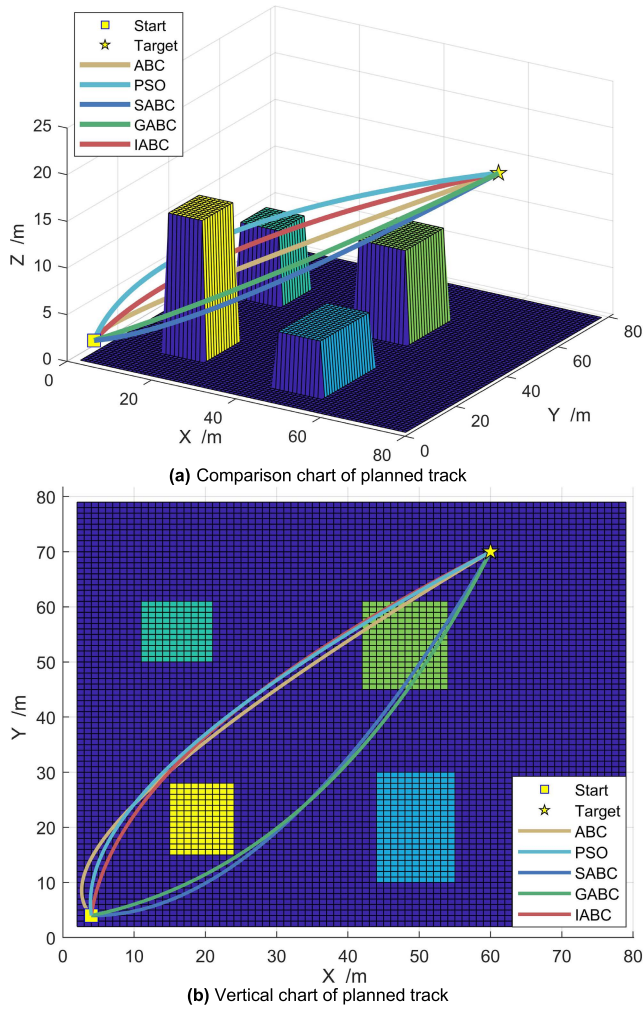
**Step 9:** Compare the individual optimal fitness value, the global optimal information updated, add 1 iteration, repeat Step4-Step8 until the number of iterations reaches  $iter_{max}$ .

**Step 10:** Complete the search and output the optimal solution.

## V. SIMULATION

In order to verify the effectiveness and feasibility of the algorithm in solving the UAV path planning problem, under the premise of ensuring objectivity and fairness, the improved artificial bee colony algorithm (IABC) and the particle swarm algorithm (PSO), the artificial bee colony algorithm (ABC), the improved artificial bee colony algorithm (SABC) proposed in [34] and the improved artificial bee colony algorithm (GABC) proposed in [35] are compared and analyzed, so that comprehensively verify the track optimization performance of the IABC algorithm.

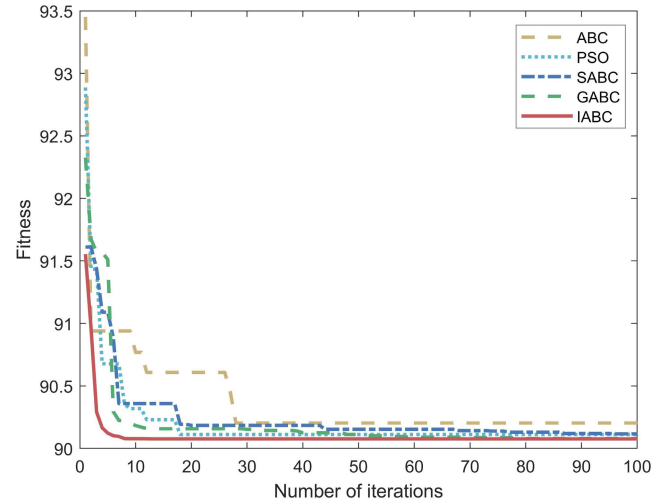
In this paper, the flight space size is  $80 \text{ m} \times 80 \text{ m} \times 25 \text{ m}$ , the starting point and end point of the mission are [4], [4], [2] and [60], [70], [15], respectively, and three environmental models are established. Detailed environmental information



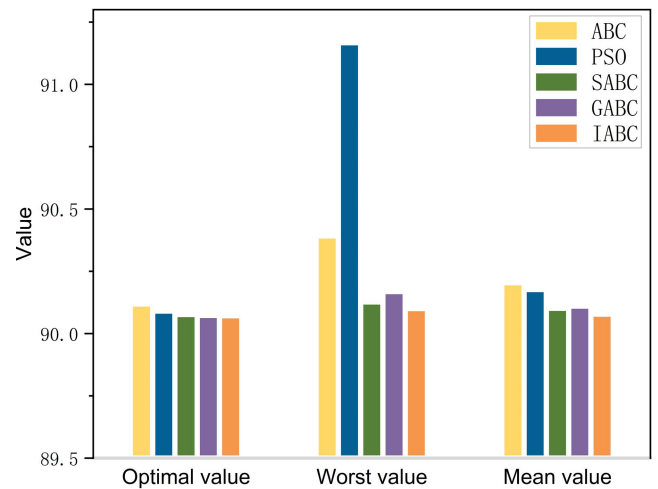
**FIGURE 7.** 3D Track planning diagram and vertical view of 5 algorithms in Case 1.

(D: depth; W: width; H: height) as shown in Table 2. The performance constraints of UAV in each simulation scenario are set to be the same (minimum flight height  $F = 5$  m, yaw angle  $\cos\theta_{\max} = 25^\circ$ , pitch angle  $\tan\beta_{\max} = 25^\circ$ ). The maximum number of iterations of all algorithms is set to  $iter_{\max} = 100$ , and each algorithm runs independently 30 times. The statistical results are shown in Table 4.

After several parameter debugging, this paper sets the IABC algorithm to guide the factor coefficient  $\psi = 2$ , which can not only guide the old solution to the global optimal solution, but also avoid falling into the local extremum trap. Set the step coefficient  $S = 4$ , so that the population can traverse the entire flight area, out of local extremum. Set the update ratio  $\omega = 1.6$ , the controllable parameter  $\sigma = 2$ , the selection of this ratio eliminates nearly 38 % of the inferior nectar sources in the population evolution, and generates a progeny bee colony with diversity and prominence. Details of the rest of this article and other algorithm parameters are listed in Table 3.

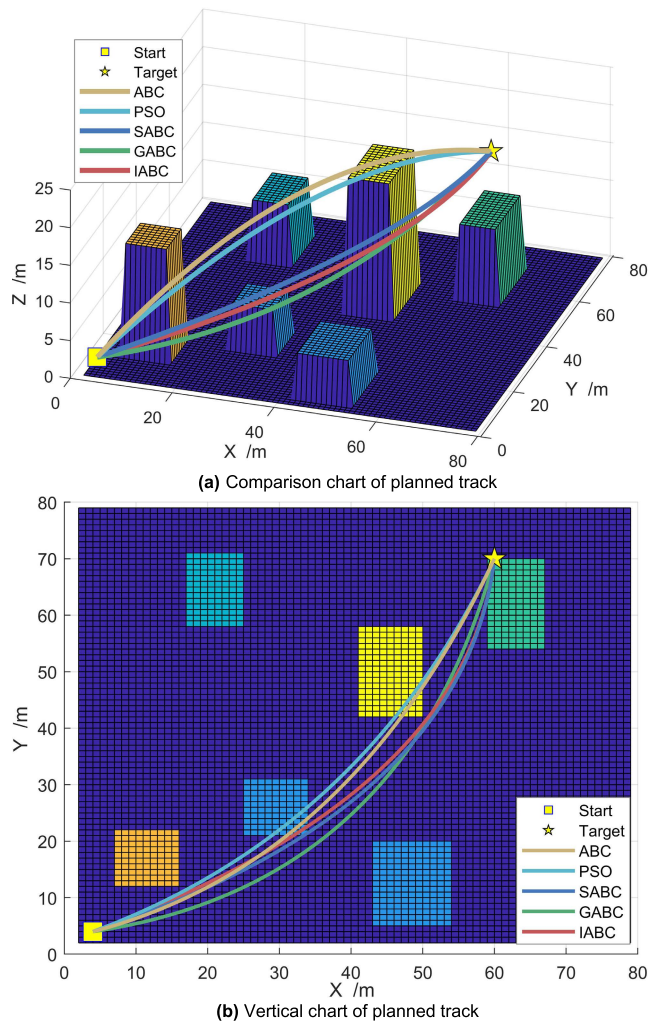


**FIGURE 8.** Iterative curve comparison of 5 algorithms in Case.



**FIGURE 9.** Statistical results of 5 algorithms in Case 1.

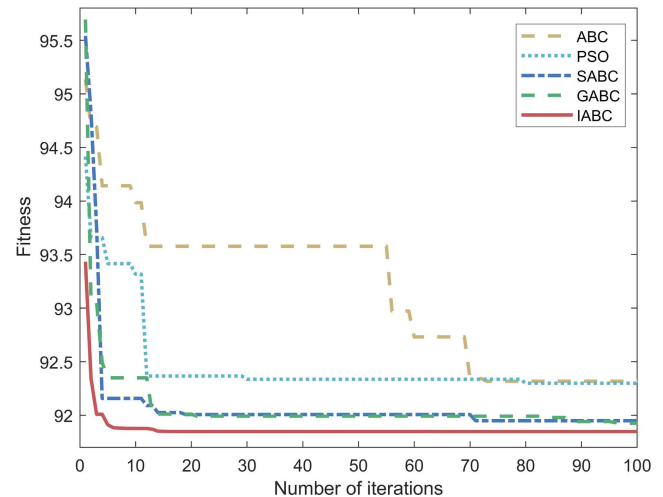
In the case 1, Fig.7 shows the different results of the five algorithms in 3D environmental trajectory planning. Although the number of obstacles in the environmental model is relatively small, according to the simulation, it can be seen that the trajectory planned by IABC meets the requirements of path planning. The flight trajectories planned by ABC, PSO, SABC and GABC generate local unnecessary trajectories at the initial stage, resulting in a longer complete trajectory distance. Fig.8 shows the convergence iteration curves of the five algorithms. It can be seen from the simulation curves that the IABC algorithm has faster iteration speed and better convergence. The IABC algorithm only needs 9 iterations to obtain the global optimal solution. The ABC, PSO, SABC and GABC algorithms need 28, 18, 87 and 80 iterations to obtain the global optimal value, respectively. The iterative curve shows that the IABC algorithm can jump out of the local extremum within 5 generations. The statistical results are shown in Fig. 9 and Table 4, ABC has a large average 90.2037, optimal value 90.1185 and worst value



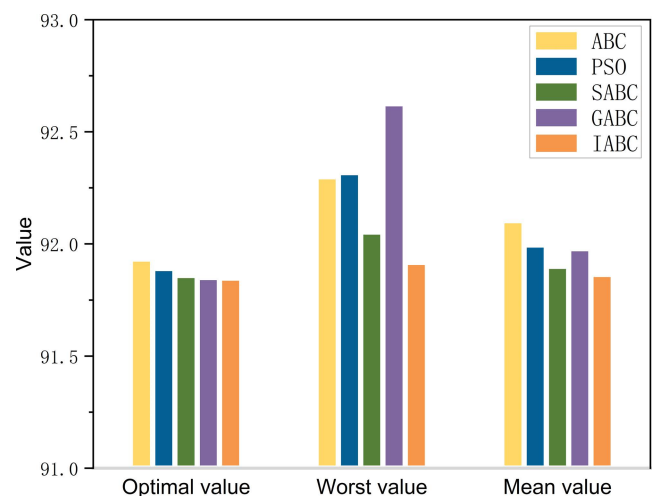
**FIGURE 10.** 3D Track planning diagram and vertical view of 5 algorithms in Case 2.

90.3917. PSO has the largest standard difference 0.1885, which means its stability is poor. The optimal value 90.0711, the inferior value 90.1001 and the mean value 90.0779 of the planned trajectory of IABC are better than those of the other four algorithms. At the same time, the low standard difference 0.0062 proves that IABC can stably search the optimal flight trajectory.

In the case 2, a more complex environmental model is designed to compare the performance of the five algorithms. As shown in Fig. 10, the difference of the flight path planned by the five algorithms is more intuitive. The flight path planned by ABC and PSO increases the flight altitude rapidly at first, and reduces the flight altitude when approaching the target point. In practical applications, such paths are not desirable, which not only affects flight safety, but also increases fuel consumption. The flight path of SABC, GABC and IABC is close, but the flight path of SABC and GABC is longer. The convergence iteration curves of the five algorithms in case 2 are shown in Fig. 11. From the simulation convergence curves, one can see that the



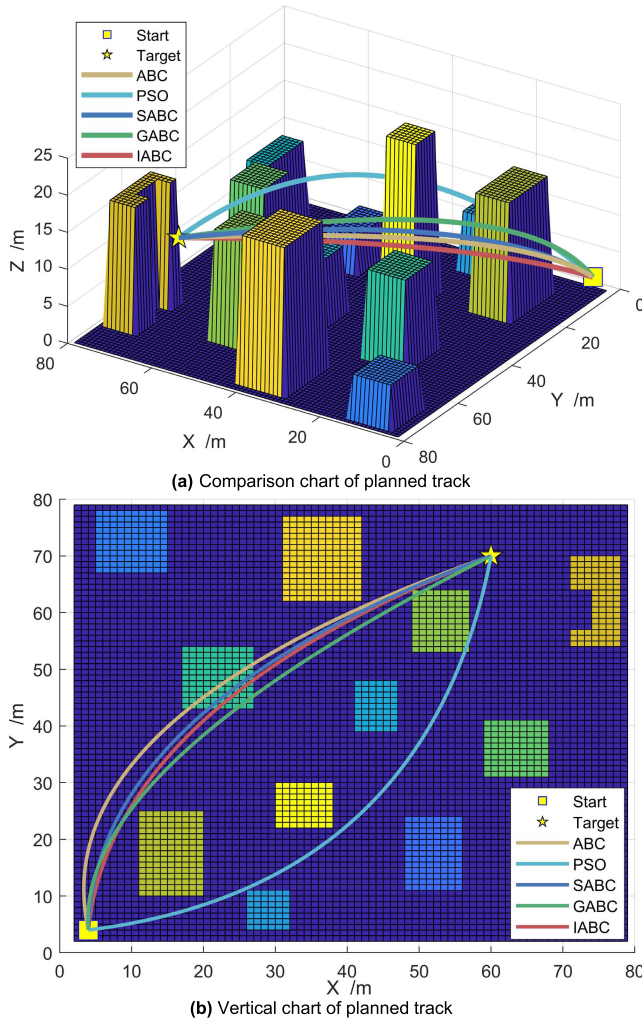
**FIGURE 11.** Iterative curve comparison of 5 algorithms in Case 2.



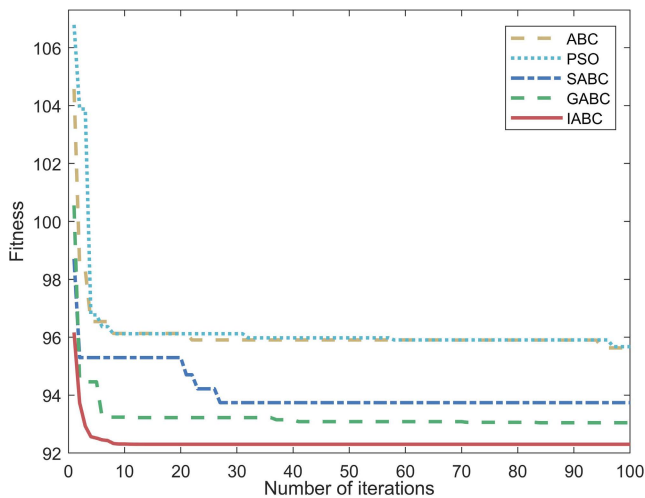
**FIGURE 12.** Statistical results of 5 algorithms in Case 2.

optimization performance of IABC is significantly better than that of the other four. IABC can solve the global optimal value only by 15 iterations. None the less, ABC needs 73 iterations, PSO needs 30 iterations, SABC needs 71 iterations, GABC needs 97 iterations. Although IABC falls into local extremum after the sixth iteration, it can get rid of local extremum after five iterations, and its efficiency is far greater than that of the unmodified ABC. The statistical results are shown in Fig. 12 and Table 4. GABC has a large standard difference 0.2174, indicating that its search stability is poor. The mean values of IABC and ABC, PSO and SABC are not significantly different, while the standard difference 0.0245 of IABC is better than that of its algorithm, the optimal, worst, mean and standard deviation of IABC are 91.8478, 91.9180, 91.8642 and 0.0245 respectively, indicating that IABC has stable planning performance and good search results.

In the case 3, we establish a more complex flight environment to test the optimization performance of the

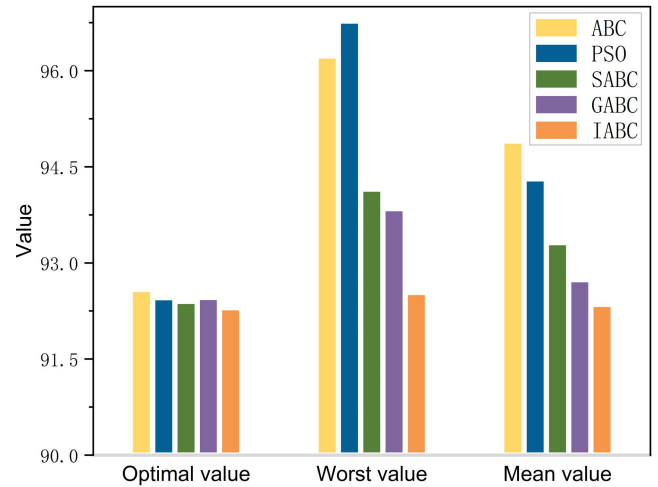


**FIGURE 13.** 3D Track planning diagram and vertical view of 5 algorithms in Case 3.



**FIGURE 14.** Iterative curve comparison of 5 algorithms in Case 3.

five algorithms. Fig.13 shows the different results of the five algorithms in 3D environmental trajectory planning.



**FIGURE 15.** Statistical results of 5 algorithms in Case 3.

**TABLE 4.** Five algorithms run statistics 30 times independently in three cases.

|        |         | ABC     | PSO     | SABC    | GABC    | IABC    |
|--------|---------|---------|---------|---------|---------|---------|
| Case 1 | Mean    | 90.2037 | 90.1762 | 90.1014 | 90.1097 | 90.0779 |
|        | Std     | 0.0613  | 0.1885  | 0.0126  | 0.0221  | 0.0062  |
|        | Worst   | 90.3917 | 91.1672 | 90.1264 | 90.1685 | 90.1001 |
|        | Optimal | 90.1185 | 90.0900 | 90.0761 | 90.0724 | 90.0711 |
| Case 2 | Mean    | 92.1037 | 91.9951 | 91.9007 | 91.9785 | 91.8642 |
|        | Std     | 0.0995  | 0.0788  | 0.0420  | 0.2174  | 0.0245  |
|        | Worst   | 92.2991 | 92.3181 | 92.0528 | 92.6253 | 91.9180 |
|        | Optimal | 91.9332 | 91.8907 | 91.8598 | 91.8505 | 91.8478 |
| Case 3 | Mean    | 94.9011 | 94.3172 | 93.3161 | 92.7369 | 92.3517 |
|        | Std     | 1.1621  | 1.2813  | 0.2134  | 0.1642  | 0.0833  |
|        | Worst   | 96.2299 | 96.7746 | 94.1513 | 93.8462 | 92.5389 |
|        | Optimal | 92.5849 | 92.4571 | 92.3990 | 92.4615 | 92.3001 |

Intuitively, the flight trajectory planned by IABC can perfectly complete the path planning task. The path distance planned by ABC and PSO is larger than that planned by IABC, which increases the fuel consumption of UAV. The flight paths planned by SABC and GABC are close to those planned by IABC, but local unnecessary trajectories are generated, which increases the complete planned path distance. In the flight trajectory of case 3, the optimization performance of IABC is better than that of the other four algorithms. Fig.14 shows the convergence curves of the five algorithms, which more obviously proves the excellent optimization performance of the IABC algorithm. It can be seen from the convergence curve that for complex environments, IABC only needs 11 iterations to obtain the global optimal solution. However, the ABC iteration is 97 times close to its global optimal value, the PSO iteration is 95 times close to its global optimal value, the SABC iteration is 27 times close to its global optimal value, and the GABC needs 84 times to obtain its optimal value. By observing Fig. 8, Fig. 11, Fig. 14, it can be concluded that IABC algorithm has high efficiency and fast convergence speed. The case 3 statistical results as shown in Fig. 15 and Table 4,

PSO has the largest standard difference 1.2813, ABC has the largest mean value 94.9011, indicating that PSO and ABC have poor stability in search. The statistical data of SABC and GABC are inferior to IABC. The optimal, worst, mean and standard deviation of IABC are 92.3001, 92.5389, 92.3517 and 0.0833 respectively. These results prove that IABC algorithm can effectively solve the UAV path planning problem in complex urban environment.

In summary, from the above experimental results, it is observed that IABC algorithm is superior to the other four algorithms in the local optimal problem, convergence speed problem and flight distance problem. We can determine that the IABC algorithm can set a safe and feasible short flight path for UAVs, and the algorithm has good stability.

## VI. CONCLUSION

In an effort to the path planning problem of UAV in urban complex environment, this paper establishes a model for UAV to seek the optimal path with multiple constraints. In order to solve this model, this paper proposes an artificial bee colony optimization algorithm based on multi-strategy comprehensive improvement, and designs the simulation comparison experiments of three different scale obstacles. The experimental results show that the improved algorithm proposed in this paper is superior to the other four comparison algorithms in stability and convergence speed, indicating that the proposed algorithm can effectively solve the UAV urban trajectory planning problem.

The artificial bee colony optimization algorithm based on multi-strategy comprehensive improvement uses the combination of Tent chaotic mapping and Pareto principle to initialize the population, which can randomly and evenly traverse the space to generate high-quality solutions and improve the operation efficiency of the algorithm. The random tangent population evolution strategy not only ensures the efficient updating of population, but also effectively avoids the generation of infeasible solutions. At the same time, the corresponding search strategy is designed to improve the convergence and diversity of the algorithm, so it can be well applied to the UAV path optimization problem with multiple constraints in complex environments. Because the encoding method and collision test of the improved algorithm are designed according to the cubic spline interpolation method, the generated track does not require curve quadratic fitting and feasibility analysis, and can directly generate the optimized track that meets the comprehensive constraints of UAV kinematics.

This algorithm has broad application prospects in trajectory planning. However, the UAV trajectory planning method studied in this paper is only applicable to the static environment, and the trajectory planning problem with the practical needs. In the future, we will focus on the multi-UAV path planning problem under dynamic threats and the adaptive adjustment of flight trajectory.

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